

CSE370 : Database Systems
Project Report
Project Title : Online Smartphone Trading Portal System

Group No : _1_, CSE370 Lab Section : _1_, Spring 2026		
ID	**Name**	**Contribution**
24101531	Maliyat Binte Marouf	
24101459	Alvi Sikder Oishe	
24101580	Sajid Ahmed	

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Introduction

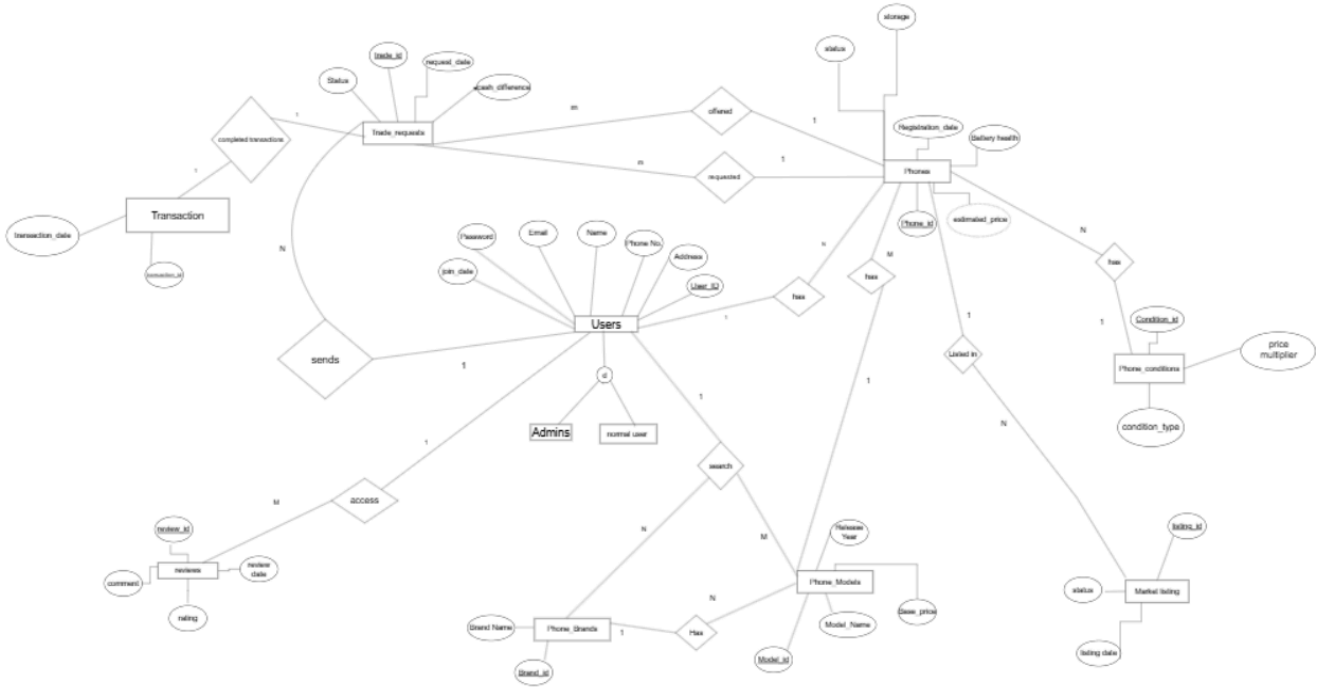
[Briefly describe the idea of your project]

This is an online smartphone trading platform which allows users to register their phone, get an estimation of the price of their phones based on factors like condition and model. Users can list their phones in a marketplace, browse other listings, and trade devices with other users. The system also includes a review-based mechanism, where users can rate and review each other after trades to ensure reliability.

Project Features

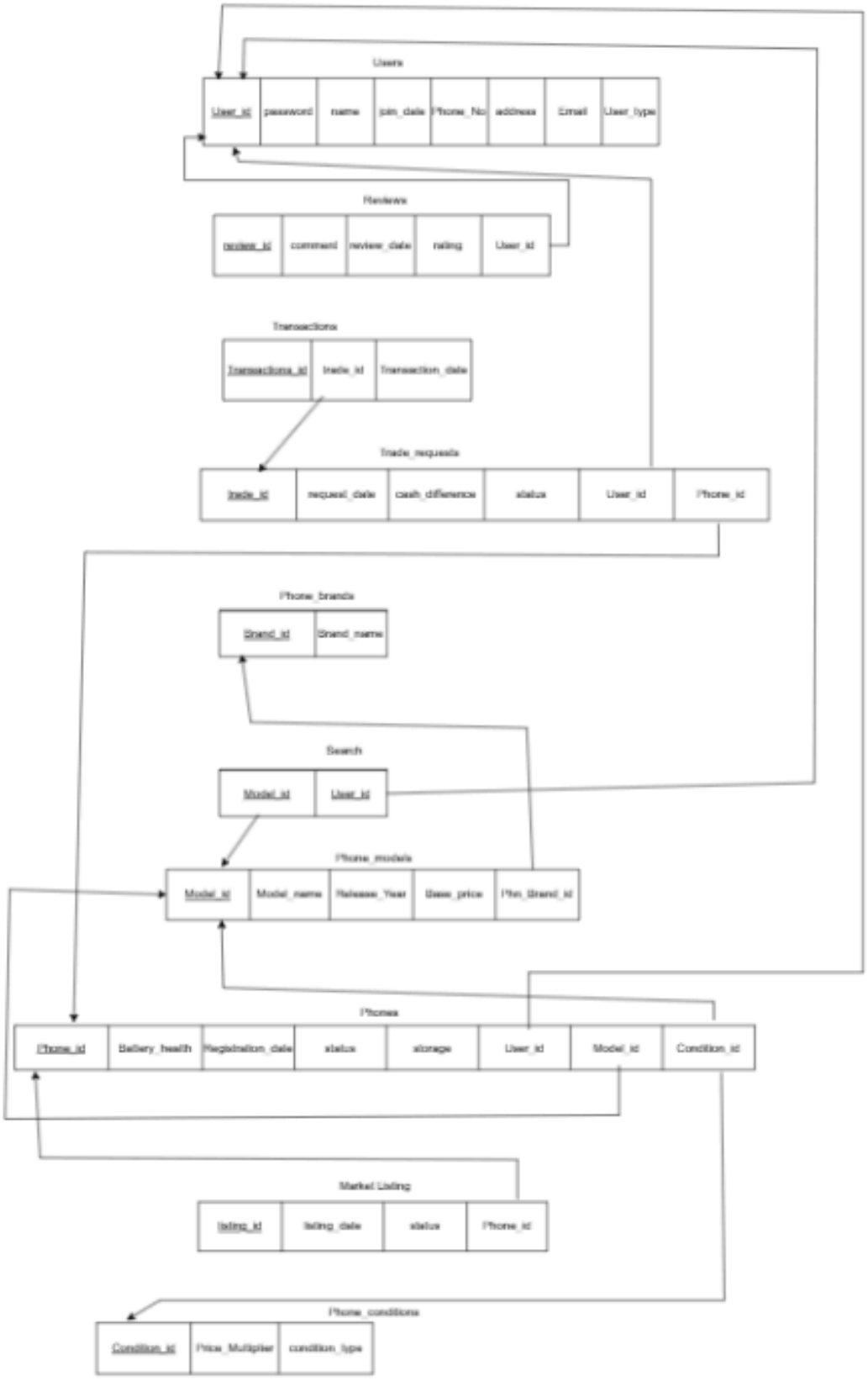
ID, Name	Features [3 per member]	
24101531, Maliyat Binte Marouf	Ft 1	Phone Valuation System
	Ft 2	User Trust & Rating System
	Ft 3	
24101580, Sajid Ahmed	Ft 1	Trade Negotiation System
	Ft 2	Trade Execution System
	Ft 3	
24101459, Alvi Sikder Oishe	Ft 1	Marketplace Discovery System
	Ft 2	Admin Monitoring & Control System
	Ft 3	

ER/EER Diagram



link : https://drive.google.com/file/d/1Vea6H5929Tznj2D4AjKDY42NYMTfPGLi/view?usp=sharing

Schema Diagram



Link:
https://drive.google.com/file/d/1ZQs-YHeshIf03qKVz7OngOCjhcFho6zV/view?usp=sharing

Normalization

a. Explain if your converted Schema is in 1NF or not. If not, decompose it to 1NF.

   Answer: The schema is already in 1NF as it does not have any composite attribute, multivalued attribute and nested relations.

b. Explain if your converted Schema is in 2NF or not. If not, decompose it to 2NF. Can there be any partial functional dependencies in your relational schema?

   Answer: The schema is already in 2NF as the schema is already in 1NF and there exists no partial dependencies.

c. Explain if your converted Schema is in 3NF or not. If not, decompose it to 3NF. Can there be any transitive dependencies in your relational schema?

   Answer: The schema is already in 3NF as the schema is already in 2NF and there exists no transitive dependencies.

Frontend Development

Briefly discuss about Backend Development and add relevant Screenshots (if required) by mentioning Individual Contributions

Basic HTML, CSS style templates and AI were implemented to create the exterior of our project.

Contribution of ID : 24101531, Name : Maliyat Binte Marouf

Contribution of ID :24101459, Name : Alvi Sikder Oishe

Contribution of ID : 24101580, Name : Sajid Ahmed

Backend Development

Briefly discuss about Backend Development and add relevant Screenshots (if required) by mentioning Individual Contributions

Contribution of ID : 24101531, Name : Maliyat Binte Marouf

price_calc.php contains a single function calculateEstimatedPrice(), which calculates a phone's estimated resale value based on three inputs - base price, release year and a condition multiplier
register.php is the frontend form page in which users fill up the information which then sends the data to register_phone.php
register_phone.php when the user submits the form register_phone.php checks if the user is logged in, if so conducts checks, gets phone info from the database which then calculates the estimated price
review.php checks if the rating provided by the user is valid (between 1 to 5), does not allow users from reviewing themselves, prevents duplicate reviews if a user has already reviewed another user they won't be allowed to leave another review and safely stores the data in the database.
Profile.php this shows the user's public profile page, shows their reviews (rating, comment), reviewer's name, shows all the phones they listed
view_phones.php this shows the list of all the phones in the platform in one table, combines 4 tables users, phones, phone_models and phone_condition

Contribution of ID :24101459, Name : Alvi Sikder Oishe

File Name	Purpose / Functionality
admin_dashboard.php	**Main admin panel with navigation options to manage users, listings, trades, reports, and phone pricing.**
admin_users.php	**Allows admins to view all users and filter them by role (admin/normal). Uses session checks and SQL filters for access control.**

admin_listings.php	**Enables admins to manage marketplace listings by activating/deactivating listings and filtering them by status through SQL updates.**
admin_models.php	**Allows updating of phone model base prices. Includes validation to prevent negative values and ensure proper SQL updates.**
admin_trades.php	**Monitors trade requests using SQL JOINs to display trade details, filter by status, and calculate cash differences.**
admin_reports.php	**Generates reports such as total users, active/inactive listings, and most listed brands using SQL COUNT() and GROUP BY queries.**
marketplace.php	**Handles marketplace operations including search, filtering, price comparison, duplicate active listing prevention, ownership validation, and dynamic SQL queries with multiple filters.**

Contribution of ID : 24101580, Name : Sajid Ahmed

File Name	Purpose / Functionality
trade.php	**Sends a trade request by allowing the user to select their phone and a target phone, while automatically calculating the cash difference between the two devices.**
my_requests.php	**Displays all sent trade requests along with their current status. Also allows users to cancel a trade request.**
incoming.php	**Shows all incoming trade requests and allows users to accept or reject them. Handles ownership transfer of traded phones, updates both phones as "Traded" in the database, removes traded phones from listings, creates transaction records, and displays full transaction history.**
transactions.php	**Displays the complete transaction history of all completed trades.**

Source Code Repository

Upload source code to GitHub or Google Drive and share a publicly accessible link in this section of the report.

Drive link :
SmartPhoneTradingPortalCode

Conclusion

References